

Lesson 3

The projection and annihilator matrices, R^2 and degrees of freedom, and the partitioned regression

1 The linear model, and the two matrices

The model is taken up again in matrix form,

$$y = X\beta + u, \quad u \sim N(0, \sigma^2 I_n),$$

with X fixed and $\text{rank}(X) = k$, so that the least-squares estimator is

$$b = (X'X)^{-1}X'y.$$

The fitted values and the residuals come from splitting y itself,

$$y = Xb + (y - Xb),$$

where the first piece is the fitted value and the second the residual:

$$Xb = X(X'X)^{-1}X'y, \quad e = y - Xb.$$

Collecting,

$$y = P_X y + (I_n - P_X)y = P_X y + M_X y,$$

where

$$P_X \equiv X(X'X)^{-1}X', \quad M_X \equiv I_n - P_X$$

are both labelled *projection matrix* in the notes: P_X projects onto the column space $L(X)$ of X , and M_X onto its orthogonal complement $L(X)^\perp$ (the residual maker, or annihilator). From here on they are also written simply P and M .

Both matrices are idempotent — and symmetric

$$P_X P_X = P_X, \quad P_X' = P_X, \quad M_X M_X = M_X, \quad M_X' = M_X.$$

For instance $P_X P_X = X(X'X)^{-1}X'X(X'X)^{-1}X' = X(X'X)^{-1}X' = P_X$, and $M_X M_X = I_n - 2P_X + P_X P_X = I_n - P_X = M_X$. A square matrix that is both idempotent and symmetric is an *orthogonal projection*.

2 How the projection and the annihilator act

The projection leaves the regressors alone, and the annihilator kills them:

$$P_X X = X, \quad M_X X = (I_n - P_X)X = X - X = 0.$$

The same pair of identities holds for the residual itself:

$$P_X e = P_X (I_n - P_X)y = 0, \quad M_X e = M_X (I_n - P_X)y = (M_X - M_X P_X)y = e,$$

the second of these because $M_X P_X = (I_n - P_X)P_X = P_X - P_X = 0$.

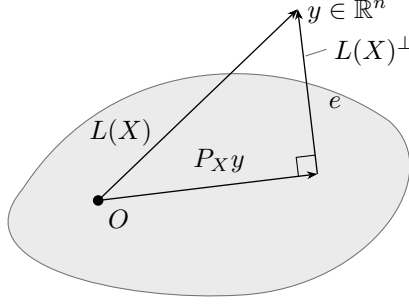


Figure 1: $P_X y$ is the projection of y onto the k -dimensional column space $L(X)$ of X , and the residual $e = y - P_X y$ is orthogonal to it. The vector in the subspace is unlabelled in the notes and is labelled $P_X y$ here. The dimensions counted next to the sketch are $\underbrace{\dim L(X)}_k + \underbrace{\dim L(X)^\perp}_{n-k} = n$.

3 The decomposition of $y'y$: explained and residual sums of squares

Inserting $y = P_X y + M_X y$ into $y'y$,

$$y'y = (P_X y + M_X y)'(P_X y + M_X y) = y'P_X y + y'M_X y,$$

the cross terms vanishing because $P_X' M_X = P_X M_X = P_X(I_n - P_X) = 0$. That is the notes' line

$$\text{T.S.S.} = \text{explained S.S.} + \text{residual S.S.},$$

and therefore

$$R^2 = \frac{y'P_X y}{y'y} = 1 - \frac{y'M_X y}{y'y}, \quad 0 \leq R^2 \leq 1.$$

This is the *uncentred* R^2 , with the total sum of squares measured from zero. When X contains a constant, the usual (centred) R^2 replaces $y'y$ by $\sum_i (y_i - \bar{y})^2$, the sum of squares around the mean, and the same decomposition holds for it. The same model written observation by observation, as on the page, is

$$y_i = x_{1i}\beta_1 + \cdots + x_{ki}\beta_k + u_i,$$

$$\vdots$$

$$y_n = x_{1n}\beta_1 + \cdots + x_{kn}\beta_k + u_n,$$

where the first subscript names the regressor and the second the observation, as the notes write it. The adjusted R^2 on the same page is

$$R_{\text{adjusted}}^2 = 1 - \frac{(y'M_X y)/(n - k)}{\sum_i (y_i - \bar{y})^2/(n - 1)},$$

each sum of squares divided by its degrees of freedom: $n - k$ for the residuals and $n - 1$ for the centred total. Unlike R^2 , it can fall when a regressor is added.

4 The expectation of the residual sum of squares

Since $M_X X = 0$, applying the annihilator to y throws away the systematic part:

$$M_X y = M_X (X\beta + u) = M_X X\beta + M_X u = M_X u,$$

so $M_X y = M_X u$, and hence $y' M_X y = u' M_X u$. The residual sum of squares is that same quadratic form in u ,

$$\text{RSS} = e'e = u' M_X u,$$

with $u \sim N(0, \sigma^2 I_n)$ written again on the page at this point. Taking expectations, and using that $u' M_X u$ is a scalar (so it equals its own trace) together with $\text{trace}(ABC) = \text{trace}(BCA)$ to turn $u' M_X u$ into $\text{trace}(M_X u u')$,

$$\begin{aligned} \mathbb{E}(e'e) &= \mathbb{E}(u' M_X u) = \mathbb{E}[\text{trace}(u' M_X u)] = \mathbb{E}[\text{trace}(M_X u u')] \\ &= \text{trace}[M_X \mathbb{E}(u u')] = \sigma^2 \text{trace}(M_X). \end{aligned}$$

The two trace facts the page records for this are

$$\text{trace}(A) = \sum_{i=1}^n a_{ii}, \quad \text{trace}(ABC) = \text{trace}(BCA) = \text{trace}(CAB).$$

It remains to evaluate the trace. Since the trace of a product may be cycled, $\text{trace}(X(X'X)^{-1}X') = \text{trace}(I_k) = k$, so

$$\begin{aligned} \text{trace}(M_X) &= \text{trace}(I_n - X(X'X)^{-1}X') \\ &= \text{trace}(I_n) - \text{trace}(I_k) = n - k, \end{aligned}$$

and therefore

$$\mathbb{E}(e'e) = \sigma^2(n - k), \quad s^2 = \frac{e'e}{n - k}, \quad \mathbb{E}(s^2) = \frac{\sigma^2(n - k)}{n - k} = \sigma^2.$$

So s^2 is unbiased for σ^2 , while $e'e/n$ is biased downwards; the k lost degrees of freedom are the k coefficients estimated. The argument used only $\mathbb{E}(u u') = \sigma^2 I_n$, not normality. Under normality, in addition, $e'e/\sigma^2 = u' M_X u/\sigma^2 \sim \chi_{n-k}^2$, since M_X is idempotent with rank = $\text{trace}(M_X) = n - k$.

5 The partitioned regression model

The regressors are now split into two blocks:

$$y = X\beta + u, \quad X = [X_1 : X_2], \quad K = k_1 + k_2,$$

with the dimensions $n \times K$, $n \times k_1$, $n \times k_2$ written under X , X_1 , X_2 . Written out,

$$y = [X_1 : X_2] \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = X_1 \beta_1 + X_2 \beta_2 + u,$$

the two coefficients being $k_1 \times 1$ and $k_2 \times 1$. The estimator expands in the same blocks:

$$b = (X'X)^{-1}X'y = [(X_1 : X_2)'(X_1 : X_2)]^{-1}(X_1 : X_2)'y = \left[\begin{pmatrix} X_1' \\ X_2' \end{pmatrix} (X_1 \ X_2) \right]^{-1} \begin{pmatrix} X_1' y \\ X_2' y \end{pmatrix},$$

and written out in full (the first line of the next page of the notes),

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} X_1' X_1 & X_1' X_2 \\ X_2' X_1 & X_2' X_2 \end{bmatrix}^{-1} \begin{bmatrix} X_1' y \\ X_2' y \end{bmatrix}.$$

6 The Frisch–Waugh–Lovell theorem: the ingredients

The notes head the next page *Frisch–Waugh–Lovell (FWL)* and set up the objects the theorem is stated in. For

$$y = X_1\beta_1 + X_2\beta_2 + u,$$

with $L(X_1)$ the column space of X_1 of dimension k_1 , the projection onto it is

$$P_1 = X_1(X_1'X_1)^{-1}X_1', \quad \dim L(X_1) = k_1, \quad \dim L(X_1)^\perp = n - k_1,$$

and the two spaces are nested — the property everything else rests on:

$$L(X_1) \subset L(X).$$

The last page of the set then collects the identities the argument runs on ($M_1 = I_n - P_1$ is the matching annihilator):

- $P_1X_1 = X_1$, $PX_1 = X_1$, $M_1X_1 = 0$;
- $M_1e_1 = e_1$, where $e_1 = M_1y$ is the residual from regressing y on X_1 alone: a residual is already annihilated, so annihilating it again changes nothing;
- $M_1M = M$, because M is the smaller of the two;
- $PP_1 = P_1$, because P_1 is the smaller of the two.

They follow from $P_1X_1 = X_1$, $PX_1 = X_1$ and idempotency. For example,

$$PP_1 = PX_1(X_1'X_1)^{-1}X_1' = X_1(X_1'X_1)^{-1}X_1' = P_1,$$

and then

$$M_1M = (I_n - P_1)(I_n - P) = I_n - P - P_1 + P_1P = I_n - P = M,$$

using $P_1P = (PP_1)' = P_1$. Symmetry gives the same identities in the other order, $P_1P = P_1$ and $MM_1 = M$. “Smaller” is what $L(X_1) \subset L(X)$ makes true: the range of P_1 lies inside that of P , and the range of M inside that of M_1 .

Transcription notes

- The set opens with a divider page carrying only “class 3”; it is not reproduced.
- The original mixes lower- and upper-case for the regressor matrix, from line to line and sometimes within a line (x on pages 2, 5, 6, 7, X on pages 4 and 8); everything is typeset with X here, including the blocks X_1 , X_2 (lower case on pages 6–9) and the column spaces $L(X)$, $L(X_1)$, as in Lesson 4. Individual elements x_{ji} stay lower case. The subscripts of the projection are written as lower-case x (P_x , M_x) on page 2 and as X elsewhere; they are typeset P_X , M_X throughout. The shorthand P , M is the notes’ own, from page 4 onwards.
- The residual maker is written in lower case (m , m_1) on pages 5, 6 and 9; it is typeset M_X , M_1 .
- The column-space glyph is the cursive lower-case letter read as ℓ in the sketch and in “ $\dim f(x)$ ”; as in Lesson 2 it is the column space of X and is typeset $L(X)$. The nested-spaces line, written “ $x_1 \subset x$ ” on page 8, is typeset $L(X_1) \subset L(X)$. In the sketch the arrow labelled $\ell(x)$ points at the shaded region (the subspace), and $\ell(x)^\perp$ at the perpendicular e ; the vector along the subspace is unlabelled.
- On page 2 the label “projection matrix” is joined by two strokes to both P_xy and M_xy ; the terms “residual maker”/“annihilator” for M_X are **Added**.

- Page 2 writes the chain $y = xb + (y - xb)$ with two down-arrows to $x(x'x)^{-1}x'y$ and to e , and the next line begins with a bare “=”; the chain is typeset continuously.
- **Corrected:** page 5 writes the chain as $My = M(x\beta + u) = Mx\beta + My = Mu$; the middle term is clearly an My (same letter shape as the y in “ $My = Mu$ ” below). It must be Mu , since $M(x\beta + u) = Mx\beta + Mu$; typeset as Mu .
- The expectation chain on page 5 repeats the argument $\text{trace}(u'Mu)$, runs into the right edge of the page, and the closing bracket of $\text{trace}(Mu u')$ is missing; it is typeset once, closed. The factor $\mathbb{E}(uu') = \sigma^2 I_n$ and the step $\text{trace}[M\mathbb{E}(uu')] = \sigma^2 \text{trace}(M)$ are written sideways in a right-hand column of the page and are folded into the chain here.
- Page 6 heads the section “Partioned Regression Model”, without the second “i”; “Partitioned” is written here. Page 8 heads the theorem “Frish-Waugh-Lovell”; “Frisch” is written here.
- On page 7 the expanded estimator carries a trailing “= 0” after the final y as written, and the last line has no leading = before its bracket; neither is reproduced. The block separators are kept as drawn: vertical dots between x_1 and x_2 , a horizontal dashed line between β_1 and β_2 .
- The row-by-row system on page 4 writes a generic row y_i (with x_{1i} , x_{ki} , u_i ; the x_{ki} could also read x_{k1}), then vertical dots, then the last row y_n ; it is typeset that way. The subscript order is regressor-then-observation, kept as written, though the observation index usually comes first.
- The adjusted R^2 is written on one line as $(y'My)/n - k$ over $y'y/n - 1$; it is typeset with the grouping that gives the usual estimator, $\text{RSS}/(n - k)$ over $\text{TSS}/(n - 1)$. **Corrected:** the $n - 1$ degrees of freedom belong to the *centred* total sum of squares $\sum_i (y_i - \bar{y})^2$ (one degree of freedom is used by $\bar{y}$), not to $y'y$; the denominator is typeset with the centred sum, and a remark that $y'P_X y/y'y$ is the uncentred R^2 is added.
- **Added:** the one-line checks of idempotency, the remarks that $\mathbb{E}(s^2) = \sigma^2$ does not need normality and that $e'e/\sigma^2 \sim \chi_{n-k}^2$ under normality, the definition of e_1 , and the derivation of $PP_1 = P_1$ and $M_1M = M$.
- **Corrected:** page 9 writes $PP_1 = P_2$ (a clear 2). Since $L(X_1) \subset L(X)$, $PP_1 = P_1$, which is also what the words on the same line (“because P_1 is smaller”) say; typeset as P_1 .
- Pages of the original, for tracing back: page 1 the divider; page 2 the model, b , the split of y into $P_X y + M_X y$, and idempotency and symmetry; page 3 $P_X X = X$, the sketch with the dimension count, and the three identities $P_X e = 0$, $M_X X = 0$, $M_X e = e$; page 4 the decomposition of $y'y$, T.S.S., R^2 and the adjusted R^2 , and the row-by-row system; page 5 $M_X y = M_X u$, $\text{RSS} = u'M_X u$, the $\mathbb{E}(e'e)$ chain and the trace identities; page 6 $\text{trace}(M_X) = n - k$, $\mathbb{E}(e'e) = \sigma^2(n - k)$, s^2 and the start of the partitioned model; page 7 the partitioned model and the expanded b ; page 8 the block formula for b , the FWL heading, $L(X_1)$, P_1 , the dimensions and $L(X_1) \subset L(X)$; page 9 the list of properties of P_1 , M_1 .